MONIRUL ISLAM

Senior Software Engineer — Backend & Platform Engineering

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SUMMARY

Senior Software Engineer with 8+ years of experience across backend and platform engineering, specializing in AWS, Kubernetes, and Infrastructure-as-Code. Led the migration of a legacy on-premise AdTech platform to AWS/EKS and GitOps-based delivery (ArgoCD), and scaled a Druid-based data warehouse to ~4x capacity. Strong background in observability, streaming data pipelines, and NestJS/Node.js backend systems across AdTech, banking, ERP, and ed-tech domains. Experienced working directly with international clients and distributed teams.

CORE SKILLS

Cloud & Infrastructure: AWS, Kubernetes (EKS), Terraform, Docker, GitOps (ArgoCD, ArgoCD Rollouts), Amazon MWAA (Airflow), ECS Fargate, KVM, Bare-metal Ops

Data & Streaming: Apache Druid, Amazon Kinesis Data Streams, Amazon Data Firehose, Vector, Fluent Bit

Observability: Prometheus, Grafana

Backend & APIs: Node.js, NestJS, REST APIs, Swagger/OpenAPI, WebSocket, Socket.io

Data Stores & Messaging: MongoDB, Redis (Cache, Pub/Sub), Oracle DB

Certifications: Databricks Certified Data Engineer, Databricks Fundamentals

EXPERIENCE

Vivasoft Limited — Jan 2024 – Present

Senior Software Engineer (L2) • *Dhaka, Bangladesh*

Client Engagement: Azerion — Bluestack (AdTech Platform)

- Led migration of a legacy on-premise AdTech platform to AWS and EKS, and transitioned Terraform-managed deployments to ArgoCD, establishing GitOps workflows with canary and blue-green delivery via ArgoCD Rollouts.
- Designed and managed AWS infrastructure with Terraform; led migration of an Apache Druid-based data warehousing platform, scaling it to ~4x its original capacity and directly supporting business growth.
- Built observability into the platform using Prometheus and Grafana, and designed log/data pipelines with Vector and Fluent Bit feeding Amazon Kinesis Data Streams and Firehose, achieving ~99% accurate data ingestion into the analytics platform.
- Owned Kubernetes cluster stability and deployment workflows, improving operational reliability for production workloads.
- Built and maintained internal developer tooling — deployment helpers, operational dashboards, and troubleshooting utilities — taking on DevOps/SRE-style ownership of day-to-day engineering operations and developer experience.
- Managed hybrid infrastructure spanning bare-metal servers, KVM virtualization, and AWS, optimizing cloud costs while maintaining scalability and reliability.
- Migrated a cron-job-based ETL pipeline for Apache Druid to AWS Managed Workflows for Apache Airflow (MWAA), with ETL jobs executed on ECS Fargate for scalable, on-demand compute.
- Mentored engineers on Node.js and backend best practices; contributed to internal tools including Vivasoft's Progress Tracker.

Vivasoft Limited — Jan 2023 – Dec 2023

Senior Software Engineer (L1) • *Banani, Dhaka*

Client Engagement: Azerion — Butelka (AdTech Platform)

- Placed as a resource-augmentation engineer with Azerion, a global AdTech company, working on Butelka — a backend platform built on NestJS and MongoDB.
- Improved backend architecture, strengthened error handling, and enhanced authentication/authorization flows to improve system reliability and security.
- Collaborated with distributed teams and international stakeholders over a 4-month engagement, delivering multiple backend improvements.
- Transitioned to the long-term Azerion Bluestack engagement after successfully completing this assignment.

APSIS — Apr 2022 – Dec 2022

Senior Software Engineer • *Banani, Dhaka*

- Built backend services for banking and supply chain ERP clients using NestJS, working directly with product owners on requirements.
- Implemented authentication with JWT and third-party SSO integration via Azure AD.
- Designed a real-time notification system using Socket.io/WebSocket and Redis Pub/Sub, plus a distributed queue management system.
- Built audit trail services, a configurable file management system, and structured application/DB logging.
- Documented APIs with Swagger/OpenAPI; reviewed code and mentored junior engineers new to NestJS, including onboarding training for new systems.

APSIS Nov 2019 – Apr 2022

Associate Software Engineer • *Dhaka, Bangladesh*

- Designed, developed, and maintained enterprise ERP modules spanning Accounts, Procurement, Sales, and Fixed Assets.
- Improved core financial reporting features (Profit & Loss, Balance Sheet, Trial Balance, Account Statements) based on evolving business requirements.
- Integrated ERP software with third-party applications and resolved production defects in collaboration with cross-functional teams.

UniSearch Dec 2021 – Mar 2022

Backend API Developer (Remote Contract) • *Australia*

- Designed and architected the backend for an ed-tech platform in NestJS, including chat/messaging and dynamic page generation.
- Built a multi-timezone appointment booking system and a virtual office feature for counsellors and students.
- Improved course search accuracy and student-to-university/course matching logic.
- Implemented Redis caching and integrated third-party weather APIs into the platform.

Earlier Career 2017 – 2019

- Software Developer, Nenorex Technologies Ltd. (2018–2019) — Automated attendance-device integration and improved payroll automation for an HRM platform.
- Software Developer, CAPM Company Ltd. (2017) — Built a company website (CodeIgniter) and a stock portfolio management application backed by Oracle DB.

EDUCATION

Jahangirnagar University 2019 – 2021

M.S. in Computer Science • *Dhaka, Bangladesh*

IBAIS University 2013 – 2017

B.Sc. in Computer Science & Engineering • *Dhaka, Bangladesh*

PERSONAL PROJECTS

Sangria — Team Chat Application sangria.chat • docs.sangria.chat

Open-source, self-hostable team communication app with channels, DMs, threads, live audio/video huddles, and polls — built end-to-end as an independent project.

Tech: *Convex, Next.js, PostgreSQL, LiveKit, Redis*

Kodama — Task Management App github.com/imonirulislam/kodama

Lightweight task management application focused on a fast, minimal workflow for individuals and small teams.

Tech: *Convex, Next.js, PostgreSQL*

Notiq — Notion-style Workspace github.com/imonirulislam/notiq

Notion-inspired notes and workspace application for organizing documents and structured content.

Tech: *Next.js, Convex*

OpenMonitor — Monitoring & Status Page github.com/imonirulislam/openmonitor

Monitoring and public status-page application for tracking service health and uptime.

Plumbline — Git Hygiene CLI github.com/imonirulislam/plumbline

Go-based CLI tool that measures repository health metrics across multiple git platforms and automatically fixes common git hygiene issues.

Tech: *Go*